الإمارات العربية المتحدة
وزارة التربية والتعليم

الميثاق الوطني للتعليم

WEEK 8 · LESSONS 2-5 · 2-6 · 3-1 · 3-2

Weekly Quiz Preparation

Grade 10 Advanced · Mathematics

كتيّب التجهّز للاختبار الأسبوعي القصير

1 2 3 4 5 6 7 8

8	10	40	24
objectives أهداف	marks علامة	minutes دقيقة	practice questions تدريبًا

Mr. Yousef Al Yousef · **Friday 23 October 2026**

One page per objective · every solution written out step by step

Quiz map

8 objectives = 8 recipes · 8 أهداف = 8 وصفات

1. **Properties of rhombi** خصائص المعين
   Diagonals bisect each other and are perpendicular · Pythagoras
2. **Tests for rhombi and squares** اختبارات المعين والمربع
   Slopes (parallel, perpendicular) · distances (equal sides) · list every name
3. **Properties of trapezoids** شبه المنحرف
   Isosceles: base angles and diagonals equal · angles on a leg add to 180°
4. **Midsegments** القطعة المتوسطة
   TS = (HJ + LK) ÷ 2
5. **Kites** الطائرة الورقية
   Angles between unequal sides are equal · angles add to 360° · perpendicular diagonals: Pythagoras
6. **The scale factor** معامل التمدد
   k = image ÷ original · k > 1 enlargement, 0 < k < 1 reduction
7. **Dilations in the plane** التمدد في المستوى
   (x, y) → (kx, ky) · plot the image and join in the same order
8. **Similar polygons** المضلعات المتشابهة
   Equal corresponding angles · proportional corresponding sides

1 · Learn
Read the idea and the recipe
اقرأ الفكرة والوصفة

2 · Watch
Cover the solution, try it, then compare
غطِّ الحل وحاول ثم قارن

3 · Try
Do “Your turn”, check at the back
حلّ التدريبات وصحّح

1 Properties of rhombi

خصائص المعين

A rhombus is a parallelogram with four equal sides — its diagonals bisect each other AND meet at a right angle.

The diagonals of a rhombus bisect each other, are perpendicular, and bisect the angles of the rhombus.

قطرا المعين ينصّف كلٌّ منهما الآخر ومتعامدان، وينصّفان زوايا المعين.

Recipe الوصفة

1. **All four sides are equal: AB = BC = CD = DA**
   الأضلاع الأربعة متساوية
2. **The diagonals bisect each other: AE = EC and BE = ED**
   القطران ينصّف أحدهما الآخر
3. **They are perpendicular: four right triangles**
   متعامدان: أربعة مثلثات قائمة
4. **Pythagoras gives a side or a half-diagonal**
   فيثاغورس لإيجاد ضلع أو نصف قطر

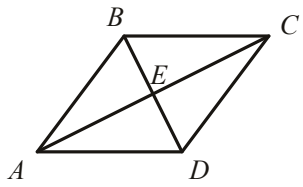
Watch out

- The diagonals of a rhombus are NOT equal (that is a rectangle).
- Half a diagonal, not the whole diagonal, is a leg of the right triangle.

Tool. $a^2 + b^2 = c^2$ in each right triangle

Worked example Authored (shape of Ex 1-8)

Quadrilateral $ABCD$ is a rhombus with diagonals meeting at E. If $AB = 13$ and $BD = 10$, find AE and AC.



1. half of the diagonal BD

$BE = \frac{1}{2}BD = 5$ (the diagonals bisect each other)

2. the diagonals are perpendicular

$\triangle ABE$ has a right angle at E

3. Pythagorean Theorem

$AE^2 = 13^2 - 5^2 = 144 \Rightarrow AE = 12$

4. the diagonals bisect each other

$AC = 2(AE) = 24$

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Ex 6) Rhombus $ABCD$, diagonals meeting at E: $AE = 4x - 3$ and $EC = 2x + 9$. Find x.
2. Authored (shape of Ex 1) Rhombus $ABCD$ with $m\angle ABD = 35°$. Find $m\angle BAC$.
3. Authored (shape of Ex 8) Rhombus $ABCD$: $AD = 3x + 5$ and $CD = 5x - 9$. Find x.

2 Tests for rhombi and squares

اختبارات المعين والمربع

Use slopes for parallel and perpendicular, distances for equal sides — then name EVERY shape that applies.

Parallelogram: opposite sides parallel. Rhombus: four equal sides. Rectangle: a right angle. Square: rhombus + rectangle.

متوازي أضلاع: ضلعان متقابلان متوازيان. معين: أضلاع متساوية. مستطيل: زاوية قائمة. مربع: معين ومستطيل معًا.

Recipe الوصفة

1. **Slopes of opposite sides: equal → parallelogram**
   ميلا الضلعين المتقابلين
2. **Distances of the four sides: all equal → rhombus**
   المسافات: كلها متساوية
3. **Consecutive slopes multiply to −1 → a right angle**
   ضرب الميلين = −1 زاوية قائمة
4. **Rhombus + right angle → square; list all that apply**
   اذكر كل ما ينطبق

Watch out

- A square is ALSO a rectangle, a rhombus and a parallelogram.
- Equal sides without a right angle: a rhombus, not a square.

Tool. slope $m = \Delta y/\Delta x$ · distance $\sqrt{(\Delta x^2 + \Delta y^2)}$

Worked example Authored (shape of Ex 17-22)

Determine whether $A(1,1)$, $B(4,5)$, $C(7,1)$, $D(4,-3)$ is a rhombus, a rectangle, a square, a parallelogram, or none. List all that apply.

1. slopes of opposite sides

$AB: \frac{4}{3}$, $DC: \frac{1+3}{7-4} = \frac{4}{3}$; $BC: -\frac{4}{3}$, $AD: \frac{-3-1}{4-1} = -\frac{4}{3}$

2. both pairs parallel

opposite sides parallel: a parallelogram

3. four equal sides

$AB = BC = CD = DA = \sqrt{3^2 + 4^2} = 5$: a rhombus

4. not a rectangle, so not a square: a parallelogram and a rhombus

$\frac{4}{3} \cdot \left(-\frac{4}{3}\right) = -\frac{16}{9} \neq -1$: no right angle

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Ex 21) $A(0,0)$, $B(3,3)$, $C(6,0)$, $D(3,-3)$: which names apply?
2. Authored (shape of Ex 20) $A(0,0)$, $B(4,0)$, $C(5,3)$, $D(1,3)$: which names apply?
3. Authored (shape of Ex 19) $A(0,0)$, $B(6,0)$, $C(6,4)$, $D(0,4)$: which names apply?

3 Properties of trapezoids

خصائص شبه المنحرف

In an isosceles trapezoid the legs are equal, the base angles are equal, and the diagonals are equal.

Base angles of an isosceles trapezoid are congruent, and so are its diagonals. Angles along a leg add to 180°.

زاويتا كل قاعدة في شبه المنحرف المتساوي الساقين متطابقتان، والقطران متطابقان، والزاويتان على الساق مجموعهما 180°.

Recipe الوصفة

1. **Find the two parallel bases**
   حدّد القاعدتين المتوازيتين
2. **Isosceles: the legs are equal, base angles equal**
   الساقان متساويتان والزاويتان متطابقتان
3. **The diagonals are equal: AC = BD**
   القطران متساويان
4. **Angles on the same leg add to 180°**
   زاويتا الساق مجموعهما 180°

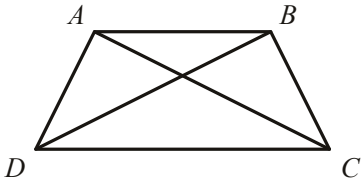
Watch out

- Base angles are on the SAME base: ∠A and ∠B for base AB.
- An ordinary trapezoid has no equal legs.

Tool. isosceles: ∠A = ∠B, ∠C = ∠D, AC = BD

Worked example Authored (shape of Ex 17-18)

$ABCD$ is an isosceles trapezoid with $\overline{AB} \parallel \overline{DC}$ and $m\angle ABC = 105°$. Find $m\angle DAB$ and $m\angle BCD$. If $AC = 5x + 1$ and $BD = 3x + 13$, find x.



1. base angles of an isosceles trapezoid
   $m\angle DAB = m\angle ABC = 105°$ (base angles)
2. consecutive angles along a leg add to 180°
   $m\angle BCD = 180 - 105 = 75°$ (angles on the leg BC)
3. the diagonals are congruent
   $AC = BD$: $5x + 1 = 3x + 13$
4. subtract 3x and 1, divide by 2
   $2x = 12 \Rightarrow x = 6$

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Ex 3) Isosceles trapezoid $ABCD$, $\overline{AB} \parallel \overline{DC}$, $m\angle A = 110°$. Find $m\angle D$.
2. Authored (shape of Ex 17) Isosceles trapezoid $ABCD$: $AC = 2x + 9$ and $BD = 4x - 5$. Find x.
3. Authored (shape of Ex 18) $ABCD$ is a trapezoid with $\overline{AB} \parallel \overline{DC}$, $m\angle A = (3x + 10)°$, $m\angle B = (5x - 30)°$. Find x so that it is isosceles.

4 Midsegments of trapezoids
القطعة المتوسطة في شبه المنحرف

The midsegment joins the midpoints of the legs; its length is the AVERAGE of the two bases.

The midsegment of a trapezoid is parallel to the bases and equals half the sum of the bases: TS = (HJ + LK) ÷ 2.

القطعة المتوسطة توازي القاعدتين وطولها نصف مجموعهما: TS = (HJ + LK) ÷ 2.

Recipe الوصفة

1. **Name the two bases (the parallel sides)**
   سمِّ القاعدتين
2. **Midsegment = (base 1 + base 2) ÷ 2**
   المتوسطة = نصف المجموع
3. **Missing base: double the midsegment, subtract the other base**
   القاعدة الناقصة = ضعف المتوسطة − الأخرى
4. **Check: the midsegment is between the two bases**
   تحقق: بين القاعدتين

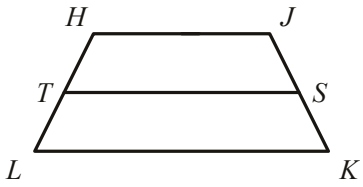
Watch out

- Divide the SUM by 2 — do not halve one base only.
- The legs are not used.

Tool. midsegment = $(b_1 + b_2) / 2$

Worked example Authored (shape of Ex 6-10)

$\overline{TS}$ **is the midsegment of trapezoid** $HJKL$**. If** $HJ = 12$ **and** $LK = 30$**, find** TS**. If** $TS = 18$ **and** $HJ = 10$**, find** LK**.**



1. half the sum of the bases

$$TS = \frac{HJ + LK}{2} = \frac{12 + 30}{2} = 21$$

2. put the known values in

$$18 = \frac{10 + LK}{2}$$

3. multiply both sides by 2

$$36 = 10 + LK$$

4. subtract 10

$$LK = 26$$

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Ex 6) $\overline{TS}$ is the midsegment. $HJ = 9$ and $LK = 25$. Find TS.
2. Authored (shape of Ex 8) $\overline{TS}$ is the midsegment. $HJ = 6$ and $TS = 14$. Find LK.
3. Authored (shape of Ex 7) The bases of a trapezoid are $2x + 1$ and $4x + 3$, and its midsegment is 14. Find x.

5 Properties of kites

خصائص الطائرة الورقية

Lesson 2-6 · Part 3 · pp. 105–106
MAT.3.09.01.019

A kite has two pairs of congruent adjacent sides; its diagonals meet at a right angle.

In a kite exactly one pair of opposite angles is congruent (the angles between the unequal sides), and the diagonals are perpendicular.

في الطائرة الورقية زاويتان متقابلتان فقط متطابقتان (بين الضلعين غير المتساويين)، وقطراها متعامدان.

Recipe الوصفة

1. **Mark the equal sides: AB = AD and CB = CD**
   علّم الأضلاع المتساوية
2. **The angles at B and D are equal; all four add to 360°**
   زاويتا B وD متساويتان ومجموع الزوايا 360°
3. **Diagonals are perpendicular: four right triangles**
   القطران متعامدان: مثلثات قائمة
4. **Pythagoras gives each side; the perimeter doubles two of them**
   فيثاغورس لكل ضلع، والمحيط = 2 × مجموع ضلعين مختلفين

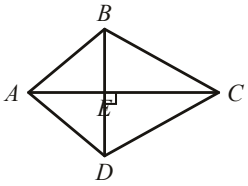
Watch out

- The congruent angles are the ones between UNEQUAL sides (B and D here).
- Round the perimeter only at the end.

Tool. m∠A + m∠B + m∠C + m∠D = 360° · $a^2 + b^2 = c^2$

Worked example Authored (shape of Ex 13-16)

Kite $ABCD$ has $AB = AD$ and $CB = CD$, with $m\angle A = 80°$ and $m\angle C = 50°$. Find $m\angle B$. The diagonals meet at E, with $AE = 12$ and $BE = 5$: find AB.



1. the angles between unequal sides are congruent
   $m\angle B = m\angle D$
2. the four angles add to 360°
   $80 + 50 + 2\,m\angle B = 360 \Rightarrow m\angle B = 115°$
3. the diagonals of a kite are perpendicular
   $\triangle AEB$ has a right angle at E
4. Pythagorean Theorem
   $AB^2 = 12^2 + 5^2 = 169 \Rightarrow AB = 13$

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Ex 13) Kite $PQRS$: $PQ = PS$, $RQ = RS$, $m\angle P = 100°$ and $m\angle R = 60°$. Find $m\angle Q$.
2. Authored (shape of Ex 14) Kite $ABCD$: $AB = AD$, $CB = CD$, $m\angle A = 120°$ and $m\angle C = 40°$. Find $m\angle D$.
3. Authored (shape of Ex 15) The diagonals of kite $ABCD$ meet at E at a right angle: $AE = 8$, $EC = 2$, $BE = ED = 6$. Find BC in simplest radical form, and the perimeter to the nearest tenth.

6 The scale factor of a dilation

معامل تمدد التحويل

Scale factor = image ÷ original. Bigger than 1: an enlargement. Between 0 and 1: a reduction.

k = (a length or coordinate of the image) ÷ (the same length or coordinate of the original). Use a vertex and its image.

k = (طول أو إحداثي الصورة) ÷ (النظير في الأصل). استخدم رأسًا وصورته.

Recipe الوصفة

1. **Read a vertex and its image from the grid**
   اقرأ رأسًا وصورته
2. **k = image ÷ original (same coordinate)**
   k = الصورة ÷ الأصل
3. **Check with a second vertex: the ratio must match**
   تحقق برأس آخر
4. **k > 1 enlargement · 0 < k < 1 reduction**
   k > 1 تكبير، 0 < k < 1 تصغير

Watch out

- Divide the IMAGE by the original, not the other way round.
- Use the same coordinate of both points (x with x, or y with y).

Tool. k = image ÷ original

Worked example Authored (shape of Ex 10, 11, 17, 18)

$\triangle A'B'C'$ is the image of $\triangle ABC$ with $A(2,1)$, $B(4,1)$, $C(3,3)$ and $A'(6,3)$, $B'(12,3)$, $C'(9,9)$. Find the scale factor.

1. a vertex and its image

   $A(2,1) \to A'(6,3)$

2. image ÷ original

   $$k = \frac{6}{2} = \frac{3}{1} = 3$$

3. the same ratio for every vertex

   $$B: \frac{12}{4} = 3, \qquad C: \frac{9}{3} = 3$$

4. larger than 1: enlargement

   $k = 3 > 1$: an enlargement

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Ex 10) $A(4,6)$ has the image $A'(2,3)$ under a dilation centered at the origin. Find the scale factor. Is it an enlargement or a reduction?
2. Authored (shape of Ex 11) A rectangle with a 6 in. side is dilated, and the matching side of the image is 15 in. Find the scale factor.
3. Authored (shape of Ex 17) $P(-3,6)$ has the image $P'(-2,4)$ under a dilation centered at the origin. Find the scale factor.

7 Dilations in the coordinate plane
التمدد في المستوى الإحداثي

Lesson 3-1 · Part 2 · pp. 120–121
MAT.3.05.05.012

A dilation centered at the origin: multiply BOTH coordinates of every vertex by k.

(x, y) → (kx, ky). Find every image vertex, plot them, and join them in the same order as the original.

(kx, ky) ← (x, y). أوجد صور الرؤوس وعيّنها ثم صلها بالترتيب نفسه.

Recipe الوصفة

1. **Write each vertex (x, y)** — اكتب كل رأس
2. **Multiply both coordinates by k** — اضرب الإحداثيين في k
3. **Plot the image vertices** — عيّن صور الرؤوس
4. **Join them in the same order** — صلها بالترتيب نفسه

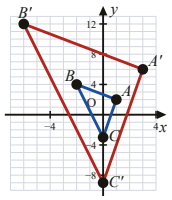
Watch out

- Multiply x AND y, never add k.
- k < 1 brings the image closer to the origin, k > 1 farther.

Tool. (x, y) → (kx, ky)

Worked example Authored (shape of Ex 13-16, 20)

$\triangle ABC$ has $A(1,2)$, $B(-2,4)$, $C(0,-3)$. Graph its image after a dilation centered at the origin with scale factor $k = 3$.



1. multiply both coordinates
   $A' = 3(1,2) = (3,6)$
2. multiply both coordinates
   $B' = 3(-2,4) = (-6,12)$
3. a zero stays zero
   $C' = 3(0,-3) = (0,-9)$
4. same order A, B, C
   plot A', B', C' and join them

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Ex 15) $P(6,-3)$, $Q(-9,12)$, $R(0,6)$, $k = \frac{1}{3}$. Find the image of $\triangle PQR$ and graph it.
2. Authored (shape of Ex 13) $J(10,-5)$, $K(-15,0)$, $L(5,20)$, $k = 0.2$. Find the image of $\triangle JKL$ and graph it.
3. Authored (shape of Ex 20) $\triangle ABC$ has $A(3,0)$, $B(-6,9)$, $C(0,-12)$. What are the coordinates of C' after a dilation centered at the origin with $k = \frac{4}{3}$? Find all three image vertices.

8 Similar polygons

المضلعات المتشابهة

Similar polygons: the corresponding angles are EQUAL and the corresponding sides are PROPORTIONAL — check both.

The similarity statement ABCD ~ WXYZ lists corresponding vertices in order: ∠A ≅ ∠W, ... and AB/WX = BC/XY = CD/YZ = DA/ZW.

عبارة التشابه ABCD ~ WXYZ تذكر الرؤوس المتناظرة بالترتيب: ∠A ≅ ∠W ... والأضلاع المتناظرة متناسبة.

Recipe الوصفة

1. **Match the vertices in the order of the statement**
   طابق الرؤوس بالترتيب
2. **Corresponding angles are congruent**
   الزوايا المتناظرة متطابقة
3. **Write the ratios of the corresponding sides**
   اكتب نسب الأضلاع المتناظرة
4. **Equal ratios and equal angles → similar; the ratio is the scale factor**
   نسب متساوية وزوايا متساوية ← متشابهان

Watch out

- Pair the sides by the STATEMENT, not by how the figures look on the page.
- Rectangles always have equal angles: only the sides decide.

Tool. corresponding sides: AB/WX = BC/XY = ...

Worked example Authored (shape of Ex 1-2, 9)

$\triangle ABC \sim \triangle DEF$. List the congruent angles and write a proportion for the sides. If $AB = 6$ and $DE = 9$, find the scale factor of $\triangle ABC$ to $\triangle DEF$.

1. the statement lists the vertices in matching order

$$\angle A \cong \angle D, \quad \angle B \cong \angle E, \quad \angle C \cong \angle F$$

2. corresponding sides are proportional

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}$$

3. image side ÷ original side

$$\text{scale factor} = \frac{DE}{AB} = \frac{9}{6} = \frac{3}{2}$$

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Ex 1) $PQRS \sim WXYZ$. Which angle is congruent to $\angle R$? Write a proportion for the sides.
2. Authored (shape of Ex 9) Rectangle $ABCD$ has sides 3 and 5; rectangle $EFGH$ has sides 9 and 12. Are they similar? Explain.
3. Authored (shape of Ex 9) Rectangle $ABCD$ has sides 4 and 6; rectangle $EFGH$ has sides 10 and 15. Are they similar? If so, find the scale factor.

Objective 1 · Properties of rhombi خصائص المعين

1.1 Authored (shape of Ex 6)

Rhombus $ABCD$, diagonals meeting at E: $AE = 4x - 3$ and $EC = 2x + 9$. Find x.

1. the diagonals bisect each other

   $AE = EC \Rightarrow 4x - 3 = 2x + 9$

2. subtract 2x, add 3, divide by 2

   $2x = 12 \Rightarrow x = 6$

1.2 Authored (shape of Ex 1)

Rhombus $ABCD$ with $m\angle ABD = 35°$. Find $m\angle BAC$.

1. rhombus property

   the diagonals are perpendicular: $m\angle AEB = 90°$

2. the angles of triangle ABE add to 180°

   $m\angle BAC = 90 - 35 = 55°$

1.3 Authored (shape of Ex 8)

Rhombus $ABCD$: $AD = 3x + 5$ and $CD = 5x - 9$. Find x.

1. all four sides are equal

   $AD = CD \Rightarrow 3x + 5 = 5x - 9$

2. subtract 3x, add 9, divide by 2

   $14 = 2x \Rightarrow x = 7$

Objective 2 · Tests for rhombi and squares اختبارات المعين والمربع

2.1 Authored (shape of Ex 21)

$A(0,0)$, $B(3,3)$, $C(6,0)$, $D(3,-3)$: which names apply?

1. slopes of opposite sides

   $AB: 1,\ DC: 1;\quad BC: -1,\ AD: -1:$ a parallelogram

2. equal sides and a right angle

   all sides $3\sqrt{2}:$ a rhombus; $1 \cdot (-1) = -1:$ a rectangle

3. rhombus + rectangle = square

   so also a square (all four names)

2.2 Authored (shape of Ex 20)

$A(0,0)$, $B(4,0)$, $C(5,3)$, $D(1,3)$: **which names apply?**

1. slopes of opposite sides

   $AB:\ 0,\ DC:\ 0;\quad BC:\ 3,\ AD:\ 3:$ a parallelogram

2. compare AB and BC

   $AB = 4,\ BC = \sqrt{10}:$ sides not all equal: not a rhombus

3. consecutive slopes

   $0 \cdot 3 \neq -1:$ no right angle: a parallelogram only

2.3 Authored (shape of Ex 19)

$A(0,0)$, $B(6,0)$, $C(6,4)$, $D(0,4)$: **which names apply?**

1. slopes: 0 and undefined

   opposite sides parallel (horizontal, vertical): a parallelogram

2. horizontal $\perp$ vertical

   all four angles are right angles: a rectangle

3. unequal adjacent sides

   $AB = 6 \neq BC = 4:$ not a rhombus, so not a square

Objective 3 · Properties of trapezoids خصائص شبه المنحرف

3.1 Authored (shape of Ex 3)

Isosceles trapezoid $ABCD$, $\overline{AB} \parallel \overline{DC}$, $m\angle A = 110°$. **Find** $m\angle D$.

1. angles on the leg AD add to 180°

   $m\angle D = 180 - 110 = 70°$

3.2 Authored (shape of Ex 17)

Isosceles trapezoid $ABCD$: $AC = 2x + 9$ **and** $BD = 4x - 5$. **Find** x.

1. the diagonals are congruent

   $AC = BD:\ 2x + 9 = 4x - 5$

2. subtract 2x, add 5, divide by 2

   $14 = 2x \Rightarrow x = 7$

3.3 Authored (shape of Ex 18)

$ABCD$ **is a trapezoid with** $\overline{AB} \parallel \overline{DC}$, $m\angle A = (3x + 10)°$, $m\angle B = (5x - 30)°$. **Find** x **so that it is isosceles.**

1. base angles of an isosceles trapezoid

   $m\angle A = m\angle B:\ 3x + 10 = 5x - 30$

2. subtract 3x, add 30, divide by 2

   $40 = 2x \Rightarrow x = 20$

Objective 4 · Midsegments of trapezoids — القطعة المتوسطة في شبه المنحرف

4.1 Authored (shape of Ex 6)

$\overline{TS}$ is the midsegment. $HJ = 9$ and $LK = 25$. Find TS.

1. half the sum of the bases

$$TS = \frac{9 + 25}{2} = 17$$

4.2 Authored (shape of Ex 8)

$\overline{TS}$ is the midsegment. $HJ = 6$ and $TS = 14$. Find LK.

1. multiply by 2

$$14 = \frac{6 + LK}{2} \Rightarrow 28 = 6 + LK$$

2. subtract 6

$$LK = 22$$

4.3 Authored (shape of Ex 7)

The bases of a trapezoid are $2x + 1$ and $4x + 3$, and its midsegment is 14. Find x.

1. half the sum of the bases

$$14 = \frac{(2x + 1) + (4x + 3)}{2} = \frac{6x + 4}{2} = 3x + 2$$

2. subtract 2, divide by 3

$$12 = 3x \Rightarrow x = 4$$

Objective 5 · Properties of kites — خصائص الطائرة الورقية

5.1 Authored (shape of Ex 13)

Kite $PQRS$: $PQ = PS$, $RQ = RS$, $m\angle P = 100°$ and $m\angle R = 60°$. Find $m\angle Q$.

1. the angles between unequal sides are congruent

$$m\angle Q = m\angle S$$

2. the four angles add to 360°

$$100 + 60 + 2\,m\angle Q = 360 \Rightarrow m\angle Q = 100°$$

5.2 Authored (shape of Ex 14)

Kite $ABCD$: $AB = AD$, $CB = CD$, $m\angle A = 120°$ and $m\angle C = 40°$. Find $m\angle D$.

1. B and D are the congruent pair

$$m\angle B = m\angle D$$

2. the four angles add to 360°

$$120 + 40 + 2\,m\angle D = 360 \Rightarrow m\angle D = 100°$$

5.3 Authored (shape of Ex 15)

The diagonals of kite $ABCD$ meet at E at a right angle: $AE = 8$, $EC = 2$, $BE = ED = 6$. Find BC in simplest radical form, and the perimeter to the nearest tenth.

1. right triangle BEC

$$BC^2 = EC^2 + BE^2 = 2^2 + 6^2 = 40 \Rightarrow BC = 2\sqrt{10}$$

2. right triangle AEB

$$AB^2 = 8^2 + 6^2 = 100 \Rightarrow AB = 10$$

3. two sides of each kind: double the sum

$$P = 2(AB + BC) = 2(10 + 2\sqrt{10}) \approx 32.6$$

Objective 6 · The scale factor of a dilation

معامل تمدد التحويل

6.1 Authored (shape of Ex 10)

$A(4, 6)$ has the image $A'(2, 3)$ under a dilation centered at the origin. Find the scale factor. Is it an enlargement or a reduction?

1. image ÷ original for x and for y

$$k = \frac{2}{4} = \frac{3}{6} = \frac{1}{2}$$

2. between 0 and 1

$0 < k < 1$: a reduction

6.2 Authored (shape of Ex 11)

A rectangle with a 6 in. side is dilated, and the matching side of the image is 15 in. Find the scale factor.

1. image side ÷ original side

$$k = \frac{15}{6} = \frac{5}{2}$$

2. larger than 1

$k > 1$: an enlargement

6.3 Authored (shape of Ex 17)

$P(-3, 6)$ has the image $P'(-2, 4)$ under a dilation centered at the origin. Find the scale factor.

1. image ÷ original

$$k = \frac{-2}{-3} = \frac{2}{3}$$

2. the y-coordinates give the same ratio

$\frac{4}{6} = \frac{2}{3}$ ✓ $0 < k < 1$: a reduction

7.1 Authored (shape of Ex 15)

$P(6,-3)$, $Q(-9,12)$, $R(0,6)$, $k=\frac{1}{3}$. **Find the image of $\triangle PQR$ and graph it.**

1. multiply both coordinates by 1/3

$P' = \frac{1}{3}(6,-3) = (2,-1)$

2. the other two vertices

$Q' = \frac{1}{3}(-9,12) = (-3,4), \quad R' = \frac{1}{3}(0,6) = (0,2)$

3. same order P, Q, R

plot and join $P'Q'R'$

7.2 Authored (shape of Ex 13)

$J(10,-5)$, $K(-15,0)$, $L(5,20)$, $k=0.2$. **Find the image of $\triangle JKL$ and graph it.**

1. multiply both coordinates by 0.2

$J' = 0.2(10,-5) = (2,-1)$

2. the other two vertices

$K' = (-3,0), \quad L' = 0.2(5,20) = (1,4)$

3. same order J, K, L

plot and join $J'K'L'$

7.3 Authored (shape of Ex 20)

$\triangle ABC$ **has** $A(3,0)$, $B(-6,9)$, $C(0,-12)$. **What are the coordinates of C' after a dilation centered at the origin with $k=\frac{4}{3}$? Find all three image vertices.**

1. multiply both coordinates by 4/3

$C' = \frac{4}{3}(0,-12) = (0,-16)$

2. the other two vertices

$A' = \frac{4}{3}(3,0) = (4,0), \quad B' = \frac{4}{3}(-6,9) = (-8,12)$

8.1 Authored (shape of Ex 1)

$PQRS \sim WXYZ$. Which angle is congruent to $\angle R$? Write a proportion for the sides.

1. R is the third vertex, so is Y

$\angle R \cong \angle Y$

2. sides in the order of the statement

$$\frac{PQ}{WX} = \frac{QR}{XY} = \frac{RS}{YZ} = \frac{SP}{ZW}$$

8.2 Authored (shape of Ex 9)

Rectangle $ABCD$ has sides 3 and 5; rectangle $EFGH$ has sides 9 and 12. Are they similar? Explain.

1. compare the corresponding sides

$\frac{9}{3} = 3$ but $\frac{12}{5} = 2.4$

2. sides not proportional

the ratios differ: not similar

8.3 Authored (shape of Ex 9)

Rectangle $ABCD$ has sides 4 and 6; rectangle $EFGH$ has sides 10 and 15. Are they similar? If so, find the scale factor.

1. corresponding sides

$\frac{10}{4} = \frac{5}{2}$, $\frac{15}{6} = \frac{5}{2}$

2. equal ratios and all right angles

equal ratios: similar, scale factor $\frac{5}{2}$

Ready?

Tick each line · ضع علامة

- [ ] **I can use the diagonals of a rhombus**
  أستطيع استخدام قطري المعين
- [ ] **I can classify a quadrilateral in the coordinate plane**
  أستطيع تصنيف الشكل الرباعي في المستوى الإحداثي
- [ ] **I can use the properties of isosceles trapezoids**
  أستطيع استخدام خصائص شبه المنحرف المتساوي الساقين
- [ ] **I can use the midsegment of a trapezoid**
  أستطيع استخدام القطعة المتوسطة
- [ ] **I can find angles and lengths in a kite**
  أستطيع إيجاد الزوايا والأطوال في الطائرة الورقية
- [ ] **I can find the scale factor of a dilation**
  أستطيع إيجاد معامل التمدد
- [ ] **I can graph the image of a polygon after a dilation**
  أستطيع رسم صورة المضلع بعد التمدد
- [ ] **I can decide whether two polygons are similar**
  أستطيع تحديد تشابه مضلعين

On the day

Draw the figure and mark what is given before you write an equation. Write the first line at once: the first step already earns a mark.

Every question

Name the property or formula you use. For a dilation, show the multiplication for every vertex.

One mark for every written step

لكل خطوة تكتبها علامة